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SPECIFICATION FOR TRANSFORMER

1. TYPE ONAN : self cooled / in / out door use / step-up / step-down / hermetically sealed distribution transformer

2. SERVICE CONDITION : The transformer will be installed at the altitudes below **1,000** meters
and the maximum ambient temperature does not exceed **40** °C

3. STANDARD : The transformer is designed, manufactured tested and tolerance in accordance with :
IEC 60076

4. RATING

4 - 1	Number of Phase	:	3	phase	4 - 2	Frequency	:	50	Hz
4 - 3	Primary Voltage	:	20	kV	4 - 4	Secondary Voltage	:	400	Volt
4 - 5	Rated Primary Current	:	28.87	A	4 - 6	Rated Secondary Current	:	1443.38	A
	Winding Connection	:	DELTA			Winding Connection	:	WYE	
4 - 7	Rated Capacity (total)	:	1,000	kVA					
	Incase of double primary voltage,					Capacity will be :			
		at	: -	kV		Rated Capacity	:	-	kVA
		at	: -	kV		Rated Capacity	:	-	kVA
	Incase of double secondary voltage,					Capacity will be :			
		at	: -	V		Rated Capacity	:	-	kVA
		at	: -	V		Rated Capacity	:	-	kVA
4 - 8	Additional Taps on High Voltage	:	+/- 2.5 % & +/- 5 % (5 Taps)						

5. POLARITY AND PHASOR RELATIONSHIP : The transformer will be subtractive polarity and phasor relationship will be : **Dyn-5**

6. ELECTRIC PERFORMANCE :

			Primary Voltage		Secondary Voltage	
6 - 1	Highest System Voltage	:	24	kV	1.1	kV
6 - 2	Impulse Test Voltage	:	125	kV	-	kV
6 - 3	Applied Test Voltage During 1 Minute	:	50	kV	3	kV
6 - 4	Induced Voltage Test :	Apply 2 times of the rate voltage at 350 cycles per second for 18 second				
6 - 5	Temperature Rise :	The winding temperature rise measured by resistance method will not exceed			65	°C
		and the oil temperature rise will not exceed			60	°C
6 - 6	Losses at Rated Voltage	Primary Side :	20	kV	Secondary Side :	0.4 kV
	No Load Losses	:	1,900	Watt		
	Load Losses (at 75 °C)	:	13,000	Watt		
	Total Losses	:	14,900	Watt		

			100	75	50	25	
6 - 7	Efficiency at % Rated Capacity	:	100	75	50	25	%
	at P.F. = 1.0	:	98.53	98.79	98.98	98.93	%
	at P.F. = 0.8	:	98.17	98.49	98.73	98.66	%

6 - 8	Exciting Current / Off load Current at Nominal Voltage	:	2.00	%
6 - 9	Percent Impedance	:	5.00	%
6 - 10	Voltage Regulation at P.F. = 1.0	:	1.42	%
	Voltage Regulation at P.F. = 0.8	:	3.98	%

To.

Date. 20 JANUARY 2021

Spec No.

7. ACCESSORIES : The transformer will be furnished with all necessary accessories as follow :

	GENERAL ACCESSORIES		SPECIAL ACCESSORIES
√	HV Porcelain Bushing on Cover	-	Plug in Bushing mm2 c/w straight connector on cover
√	LV Porcelain Bushing on Cover	√	Integrated Safety Device c/w Contact
-	Oil Level Indicator w/o Contact	-	Oil Level Indicator c/w Alarm Contact
-	Oil Temperature Indicator w/o Contact	-	Oil Temperature Indicator c/w Alarm & Trip Contact
-	Pressure Relief Valve w/o Contact	-	Pressure Relief Device c/w Alarm Contact
√	Name Plate	-	Winding Temperature Indicator c/w Alarm & Trip Contact
√	Earth Terminal	-	CT for Winding Temperature Indicator
√	Manual Off-Load Tap Changer	-	Sudden Pressure Rise Relay + Seal in Relay c/w Alarm & Trip Contact
-	Voltage Change Over Board	-	Buchholz Relay c/w Alarm & Trip Contact
√	Radiator	-	Conservator c/w Sampling Valve
√	Lifting Lug	-	Breather c/w Silicagel
√	Oil Filter Valve	-	Valve for Buchholz Relay
√	Drain Valve	-	Pressure Vacuum Gauge c/w Bleeder
√	Roller	-	N2 Filling Valve
-	Skid Base (Pad Mounted)	-	Air-filled LV Junction Box
-	Jacking Lug	-	HV/LV Cable Box
-	Sampling Valve	-	Terminal Control Box
-	Ladder	-	Radiator Fans
		-	Radiator Valves
		-	Neutral CT
		-	Thermometer Pocket

No.	**Routine Test**
1	*Breakdown voltage oil*
2	*Transformer Turn Ratio, Polarity & Vector group test*
3	*Measurement of No load loss and Exciting current*
4	*Measurement of impedance and load loss*
5	*Separate source voltage withstand test (Applied potential test)*
6	*Induced potential test*
7	*Insulation resistance test*
8	*Winding resistance measurement*

* Painting System Stoving Min. 70 Micron
* Painting Color RAL 7023 (Concrete Grey)

Note :

PROPOSER :	Date of Approval :
hp	